

# Glossary

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**Annuity ( $A(t)$ )**

The sum of accrual-weighted discount factors across a swap's payment dates; converts a swap-rate quantity into money. Also called the PVBP or (up to scaling) the DV01.

**Bachelier's formula**

The option pricing formula obtained by assuming the underlying rate follows a driftless normal (arithmetic) process; handles negative rates without difficulty.

**BGM model**

The Brace–Gatarek–Musielà model, also called the LIBOR Market Model; models discrete, market-observable forward rates directly as lognormal (or displaced-lognormal) martingales under their own forward measures.

**Black's formula**

The option pricing formula obtained by assuming the underlying rate follows a driftless lognormal (geometric Brownian) process; undefined for a non-positive underlying.

**Cap**

A strip of caplets sharing a common strike, protecting the holder against a floating rate rising above that strike.

**Caplet**

A single-period call option on a forward rate; the building block of a cap.

**Cascade calibration**

A sequential BGM calibration procedure that walks through co-terminal swaptions in a fixed order, solving (approximately) one new unknown parameter at each step.

**Change of numeraire**

The technique of re-expressing asset prices in units of a different tradable asset (numeraire), which changes the associated pricing measure and the drift, but not the volatility, of underlying processes.

**Compounded-in-arrears rate**

A backward-looking interest rate constructed by compounding daily overnight fixings across an accrual period; known only at the period's end. The dominant SOFR cap/floor convention.

**Discount factor ( $P(t, T)$ )**

The value at time  $t$  of \$1 received with certainty at time  $T$ .

**Displaced diffusion**

A BGM extension in which the rate's volatility is applied to  $(L_i + c_i)$  rather than  $L_i$  alone, for a displacement  $c_i \geq 0$ , lowering the model's negative-rate floor and generating skew.

**Forward measure**

The pricing measure associated with a zero-coupon bond numeraire, under which the corresponding simple forward rate is a driftless martingale.

**Girsanov's theorem**

The result describing how a Brownian motion's drift transforms under a change of measure; the diffusion coefficient is unaffected.

**HJM framework**

The Heath–Jarrow–Morton framework, modelling the entire instantaneous forward curve directly, with drift determined by volatility via the HJM no-arbitrage condition.

**Ill-posedness (calibration)**

A situation in which many materially different parameter vectors fit observed market data almost equally well, making the fitted parameters individually unreliable even when the overall fit quality is good.

**Normal (Bachelier) volatility ( $\sigma_n$ )**

Volatility expressed in the rate's own absolute units (e.g. bp/yr), independent of the rate's level.

**Lognormal (Black) volatility ( $\sigma$ )**

Volatility expressed as a percentage of the current rate level.

**Predictor-corrector scheme**

A Monte Carlo discretisation of BGM that evaluates the state-dependent drift twice per time step (predicted, then corrected using the predicted rate levels) and averages the two, for second-order accuracy.

**Rank reduction**

Approximating a full-rank correlation matrix by a lower-rank matrix built from its largest eigenvalues, reducing the number of independent Brownian factors needed in simulation.

**Rebonato's swaption approximation**

A formula approximating a swap rate's volatility as a frozen-weight combination of the volatilities and correlations of its underlying forward rates; the standard bridge between BGM parameters and swaption prices.

**SABR model**

Hagan et al.'s stochastic-volatility model (Stochastic Alpha, Beta, Rho), the market-standard tool for describing and interpolating a rates volatility smile.

**SOFR**

The Secured Overnight Financing Rate; the primary USD risk-free rate that replaced LIBOR.

**Spot (caplet) volatility**

The volatility applicable to a single individual caplet, as distinct from a cap's flat quoted volatility.

**Swaption**

An option to enter an interest rate swap at a future date; a payer swaption is a call on the forward swap rate.

**Swaption cube**

The three-dimensional (expiry  $\times$  tenor  $\times$  strike) swaption volatility surface.

**Terminal correlation**

The realised correlation between two rates' values at a future date, as distinct from their instantaneous correlation at a single point in time; the deeper quantity the frozen-weights swaption approximation approximates.

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**Terminal measure**

The pricing measure associated with the final bond  $P(t, T_N)$  in a tenor structure.



# Formula Reference

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## Curves and rates (Chapter 2)

Discount factor  $P(t, T), \quad P(T, T) = 1$

Simple forward rate  $L(t; T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left( \frac{P(t, T_1)}{P(t, T_2)} - 1 \right)$

## Swaps (Chapter 4)

Annuity  $A(t) = \sum_{i=1}^n \tau_i P(t, T_i)$

Par swap rate  $S(t) = \frac{P(t, T_0) - P(t, T_n)}{A(t)}$

## Caplets (Chapter 6)

Black  $N\tau P(0, T_2) [F\Phi(d_1) - K\Phi(d_2)], \quad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2 T_1}{\sigma\sqrt{T_1}}$

Bachelier  $N\tau P(0, T_2) \sigma_n \sqrt{T_1} [d\Phi(d) + \varphi(d)], \quad d = \frac{F - K}{\sigma_n \sqrt{T_1}}$

Cap–floor parity  $\text{Caplet} - \text{Floorlet} = N\tau P(0, T_2)(F - K)$

## Swaptions (Chapter 7)

Black  $NA(0) [S\Phi(d_1) - K\Phi(d_2)]$

Bachelier  $NA(0) \sigma_n \sqrt{T} [d\Phi(d) + \varphi(d)]$

ATM shortcut  $NA(0) \sigma_n \sqrt{T/2\pi}$

## HJM (Chapter 10)

Drift condition  $\alpha(t, T) = \sigma(t, T) \int_t^T \sigma(t, u) du$

**BGM (Chapters 11, 13)**

Own-measure dynamics  $dL_i(t) = \sigma_i(t)L_i(t) dW_i^{i+1}(t)$

Numerator-step vol  $\gamma_j(t) = \frac{\tau_j \sigma_j(t) L_j(t)}{1 + \tau_j L_j(t)}$

Terminal-measure drift  $-\sigma_i(t)L_i(t) \sum_{j=i+1}^{N-1} \tau_j \rho_{ij} \sigma_j(t) \frac{L_j(t)}{1 + \tau_j L_j(t)}$

Spot-measure drift  $+\sigma_i(t)L_i(t) \sum_{j=\eta(t)}^i \tau_j \rho_{ij} \sigma_j(t) \frac{L_j(t)}{1 + \tau_j L_j(t)}$

**Volatility, correlation, smile (Chapters 14–15)**

Rebonato hump  $\sigma_i(t) = \phi_i[(a + b(T_i - t))e^{-c(T_i - t)} + d]$

Correlation form  $\rho_{ij} = \rho_\infty + (1 - \rho_\infty)e^{-\beta|T_i - T_j|}$

Displaced diffusion  $dL_i(t) = \sigma_i(t)(L_i(t) + c_i) dW_i^{i+1}(t)$

**Calibration (Chapters 17–18)**

Rebonato swaption approx.  $(\sigma^{\text{swaption}})^{2T} \approx \frac{1}{S(0)^2} \sum_{i,j} w_i(0)w_j(0)L_i(0)L_j(0)\rho_{ij} \int_0^T \sigma_i \sigma_j dt$

Joint objective  $\theta^* = \arg \min_{\theta} \sum_k w_k (\sigma_k^{\text{mkt}} - \sigma_k^{\text{model}}(\theta))^2 + \lambda R(\theta)$

# Bibliography and Further Reading

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## Primary references

**Brace, A., Gatarek, D. and Musiela, M. (1997)**

*The Market Model of Interest Rate Dynamics*, Mathematical Finance 7(2), 127–154. The founding BGM paper — source of the direct-modelling-of-simple-forward-rates construction that gives this book its subject.

**Miltersen, K., Sandmann, K. and Sondermann, D. (1997)**

*Closed Form Solutions for Term Structure Derivatives with Log-Normal Interest Rates*, Journal of Finance 52(1), 409–430. Independently published the same core idea the same year; jointly credited with Brace–Gatarek–Musiela as the origin of the market model.

**Heath, D., Jarrow, R. and Morton, A. (1992)**

*Bond Pricing and the Term Structure of Interest Rates: A New Methodology for Contingent Claims Valuation*, Econometrica 60(1), 77–105. Source of the HJM no-arbitrage drift condition built on in Chapter 10, and the direct intellectual ancestor of the BGM construction.

**Hagan, P., Kumar, D., Lesniewski, A. and Woodward, D. (2002)**

*Managing Smile Risk*, Wilmott Magazine. Source of the SABR model and its closed-form implied volatility approximation, used throughout Chapter 8 and Chapter 15.

**Jamshidian, F. (1997)**

*LIBOR and Swap Market Models and Measures*, Finance and Stochastics 1(4), 293–330. Companion treatment of the market model, with particular focus on the swap market model and change-of-numeraire technique used throughout Chapter 12.

**Rebonato, R. (2002)**

*Modern Pricing of Interest-Rate Derivatives: The LIBOR Market Model and Beyond*. Princeton University Press. Source of the humped volatility parameterization (Chapter 14) and the frozen-weights swaption approximation (Chapter 17) used throughout this book's calibration chapters.

**Brigo, D. and Mercurio, F. (2006)**

*Interest Rate Models — Theory and Practice*, 2nd ed. Springer Finance. The standard comprehensive reference for the full range of term-structure models discussed in Part III, and for the multi-curve extensions discussed in Chapter 3.

**Hunter, C., Jäckel, P. and Joshi, M. (2001)**

*Drift Approximations in a Forward-Rate-Based LIBOR Market Model*, Getting the Most from Foreign Exchange (Risk Publications). Source of the predictor-corrector Monte Carlo scheme built in Chapter 19.

**Alexander, C. (2002–2003)**

*Common Correlation Structures for Calibrating the LIBOR Model*, working paper series. Source for the parametric correlation forms and rank-reduction techniques of Chapter 14.

**ARRC (Alternative Reference Rates Committee)**

Various publications, 2017–2023, including the recommended conventions for SOFR-based

cash products referenced in Chapter 3. [newyorkfed.org/arrc](http://newyorkfed.org/arrc).

### Caveats on the material in this book

- **Market conventions move.** This book was written with current, mid-2026 SOFR-market conventions in mind (Chapter 3). Rates market infrastructure has changed substantially before (the LIBOR transition itself) and individual conventions — day-count bases, settlement mechanics, which tenor of a term rate is most liquid — can and do continue to shift. Verify any specific convention against a current primary source (ISDA definitions, ARRC publications, or a live market data vendor) before relying on it in a system that risks capital.
- **Numerical examples use indicative, not live, market levels.** Every worked example in this book uses plausible, realistic 2025–2026-era SOFR-market parameter values, chosen to illustrate the arithmetic clearly and to be internally consistent within each example. None of these numbers should be treated as an actual market quote at any specific date; always source live parameters from a market data feed before pricing real risk.
- **The Rebonato swaption approximation is an approximation.** Chapter 17 is explicit about this, but it bears repeating in the back matter: the frozen-weights formula is a widely used and generally accurate practitioner tool, not an exact result, and its accuracy degrades specifically for long-expiry, long-tenor swaptions. A production system should validate the approximation's accuracy against full Monte Carlo repricing (Chapter 19) periodically, not assume it holds uniformly well across the entire cube.
- **This book does not cover every extension in current use.** Notably out of scope: full stochastic-volatility SABR–LMM calibration mechanics in complete detail (Chapter 15 sketches the model but does not derive its calibration in the depth given to plain displaced-diffusion BGM); multi-currency and cross-currency BGM extensions; credit and funding valuation adjustments (CVA/FVA) layered on top of a BGM pricing engine; and the full mathematical machinery of Bermudan swaption exercise boundary determination (Longstaff–Schwartz regression and its variants), which is referenced in Chapter 19 but not built from scratch. Each is a substantial topic capable of supporting its own book-length treatment.
- **What is contested versus settled.** The core no-arbitrage mathematics of Parts III and IV (the HJM and BGM drift conditions) is settled, peer-reviewed theory with decades of scrutiny. The specific parametric forms in Chapter 14, the choice between cascade and joint calibration in Chapter 18, and the specific governance and architecture recommendations of Chapter 20 reflect widely used, well-established practitioner consensus, but are engineering and risk-management choices rather than mathematical theorems — reasonable, well-run desks differ in the specific details of how they implement them.